EVEI LIGHT METER — Arduino Mega 2560 carrier shield

microSD on board socket, 3.3V LDO, 74LVC125 shifts MOSI/SCK/CS 5V->3.3V; MISO (3.3V) goes straight to D50 (same as Adafruit 254 breakout).

Pin map from latest_version_of_lightmeter.ino: SD CS=D53 (HW SPI 50/51/52), GPS=Serial1 (D18/D19), IMU=Serial2 (D16/D17), I2C=D20/D21,
LEDs common-anode to 5V, active LOW. setLED(GREEN) pulls the *_red pin LOW, so *_red pins drive the GREEN die: GPS G=D4 R=D7, LUX G=D5 R=D6, IMU G=D8 R=D9, LUX D5/D6, IMU D8/D9; buttons to GND with INPUT_PULLUP; sleep D10, log toggle D11, force log D12.

Connector pinouts (JST-XH): LCD/LUX = GND,5V,SDA,SCL LEDx = 5V(common anode),RED,GREEN BTNx = GND,5V,SIG(pull-up)

REV B ADDITIONS

Second TCA9548A (U4, addr 0x71) on J10: fixed I2C address 0x29 clashes with LUX1, so it sits behind PCA9548A channel 0.
Mux powers up with all channels off, so LUX1 on the main bus works with unchanged firmware. To read LUX2: write 0x01 to 0x70, read, write

Encoder (J11): A -> D2 (INT4), B -> D3 (INT5). 10k pull-ups for open-collector encoders, 1k + 1nF filter and pin protection.
GPS LED moved from D2/D3 to D4/D7 to free both interrupt pins: change gps_red = 4, gps_green = 7 in the sketch.

IMU in UART mode: PS1 tied to breakout 3Vo via JP1. BNO SDA pin = UART TX -> Mega RX2 (D17): BNO SCL pin = UART RX -> Mega TX2 (D16).

Purdue EVEI	
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